

From Coccyx Pain to Neurological Impairment: Investigating a Case of Pachymeningeal Enhancement



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INTRODUCTION

Pachymeningeal enhancement is a rare imaging finding indicative of inflammation or infiltration of the dura mater. It is often associated with a broad range of underlying conditions, including infectious, autoimmune, neoplastic, and idiopathic processes. Presenting symptoms can vary widely, often posing significant diagnostic challenges. This case illustrates an unusual initial presentation with coccyx pain and progressive neurological deficits in a young woman.

CASE PRESENTATION

A 35-year-old woman with a PMHx of PCOS and HLD presented with a one-month history of progressive coccyx pain and constipation, initially attributed to coccydynia following a vaginal delivery three months prior. She eventually returned presenting with severe sepsis and significant bilateral lower extremity weakness and a new inability to ambulate. A pan-CT revealed no evidence of infection or malignancy. However, MRI revealed diffuse pachymeningeal and leptomeningeal enhancement extending from T8 to the sacrum. A lumbar puncture was attempted but yielded only 2 mL of cerebrospinal fluid (CSF) due to diffuse leptomeningeal thickening. During the hospital course, the patient developed tender, erythematous nodules on her lower extremities, consistent with erythema nodosum.

MIND-V Category	Findings in This Case	Conclusion
M — Malignancy	Pan-CT negative for neoplasm; no evidence of malignancy on MRI.	Malignancy unlikely
I — Infection	CSF viral panel, CSF culture, TB, and Histo all negative; lymphocytic pleocytosis	Infection unlikely
N — Non-infectious Inflammatory	Elevated ANA (1:1280) with elevated ESR and CRP RF/Anti-CCP, ANCA (PR3/MPO), Anti-dsDNA, Anti-Sm, SSA/SSB Negative Serum IgG4 — Normal Serum ACE — Normal	Autoimmune process suspected; narrowed to IgG4-RD and neurosarcoidosis
D — Drug-Related	No history of intrathecal drugs or other medications known to cause pachymeningitis.	Drug-induced unlikely
V — Vascular	Normal opening pressure and normal closing pressure on LP; no signs of low ICP.	Low ICP unlikely

Table 1: Systematic evaluation of pachymeningitis using the MIND-V framework



Figure 1 T2-weighted sagittal MRI of the cervical, thoracic, and lumbar spine (from left to right)

DIAGNOSTIC CHALLENGES

- **Neurosarcoidosis** is a rare manifestation of sarcoidosis, characterized by **granulomatous** inflammation of the nervous system, commonly causing **cranial neuropathies, pachymeningitis, and hydrocephalus**
- **IgG4-RD** is a systemic immune-mediated condition characterized by fibroinflammatory processes that **can affect nearly any organ**, typically involving the pancreas, bile ducts, salivary glands, kidneys, or retroperitoneal structures
- While the **normal serum ACE level and negative chest imaging** do not support pulmonary sarcoidosis, these findings **do not exclude isolated neurosarcoidosis**, as ACE sensitivity is variable and non-pulmonary involvement is common
- **Serum IgG4 is elevated in approximately two-thirds of cases**, though it is neither sensitive nor specific on its own
- A **leptomeningeal biopsy is the gold standard** for diagnosing both neurosarcoidosis (via confirmation of non-caseating granulomas) and IgG4-related disease (via identification of IgG4-positive plasma cells, storiform fibrosis, and obliterative phlebitis)
- Both neurosarcoidosis and IgG4-RD are steroid-responsive conditions, but **long-term management should focus on steroid-sparing agents like mycophenolate or methotrexate**

DISCUSSION

The primary differential diagnoses for this patient’s presentation are neurosarcoidosis or IgG4-related disease (IgG4-RD). The initial presentation of coccyx pain in this patient, while unusual, could represent an early manifestation of pachymeningeal inflammation. Coccyx pain may have resulted from irritation or involvement of the sacral nerve roots, as diffuse leptomeningeal thickening extended from T8 to the sacral region on MRI. The diagnostic workup in this case was complicated by the technical challenges of obtaining cerebrospinal fluid (CSF) due to diffuse thickening of the leptomeninges. While a limited CSF sample was obtained, it was insufficient to perform a comprehensive analysis, highlighting a diagnostic barrier in patients with hypertrophic pachymeningitis. In such cases, alternative approaches, such as fluoroscopic-guided lumbar puncture or repeat attempts after steroid treatment, could improve diagnostic yield by alleviating inflammation and thickening. Given the limitations of CSF studies, early consideration of a leptomeningeal biopsy may have been warranted to establish a definitive diagnosis. In the absence of sufficient CSF analysis, the utility of emerging biomarkers becomes increasingly relevant. Emerging biomarkers for neurosarcoidosis include soluble IL-2 receptor (sIL-2R), which correlates with disease activity and can guide diagnosis when sarcoidosis is suspected. For IgG4-RD, elevated IgG4:IgG ratio and serum eosinophil counts are being explored as potential markers to support diagnosis in challenging cases.

CONCLUSION

Aside from being a valuable teaching case, this case demonstrates the diagnostic limitations in neurosarcoidosis and IgG4-RD, particularly when invasive testing is not feasible. Advanced diagnostics, such as next-generation sequencing of cerebrospinal fluid or serum-based assays for inflammatory cytokines, may offer a non-invasive alternative to biopsies in the future.

REFERENCES

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